

Project Proposal

Group Members

Francesco Macrì – Student ID 2055851

Ventrone Lorenzo – Student ID 1802393

Project Type

Comparison between a Relational DBMS (PostgreSQL) and a Graph Database (Neo4j).

Dataset Description

The **IMDb Non-Commercial Dataset** is a freely available collection of tab-separated files published by the Internet Movie Database, containing metadata about movies, TV productions, cast and crew assignments, and user-aggregated ratings. The relevant files are `title.basics`, `name.basics`, `title.principals`, `title.crew`, and `title.ratings`. To keep the scope manageable, the data will be filtered to feature films released after 1990 with at least 1,000 votes (~ 200000 titles, ~ 3000000 person–title relationships).

Reference: <https://datasets.imdbws.com/>

Project Description

The project implements the same set of analytical queries over the IMDb dataset using both PostgreSQL (relational) and Neo4j (graph), then compares the two approaches in terms of query expressiveness, readability, and execution performance. The dataset will be modelled relationally (normalised tables with foreign keys) and as a property graph (nodes for **Title**, **Person**, **Genre**; edges for **ACTED_IN**, **DIRECTED**, **HAS_GENRE**). Target queries include: shortest collaboration path between two actors (*Six Degrees of Separation*), identification of the most connected actors in the collaboration network, genre-based rating aggregations by decade, and content-based movie recommendations via shared cast and crew. The outcome will highlight where graph traversal (Cypher `shortestPath`) offers a clear advantage over equivalent recursive SQL, and where relational aggregation remains more natural and performant.